NEXUS: Navigating Emergent X-agent Universe Simulator with Unprecedented Insight

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Abstract

Conventional public opinion analysis systems face challenges in integrating fragmented, high-noise, cross-platform information streams, especially regarding the synergistic modeling of dynamic information retrieval and social stimulation. To address these limitations, this paper proposes the NEXUS framework. It deeply integrates four core modules: 1) a domain-adapted fine-tuned LLM, 2) user-anchor-guided deep search, 3) semantic-graph knowledge structuring, and 4) multi-agent virtual society simulation, forming a complete end-to-end Retrieval-Simulation-Decision analytical pipeline. Experimental results demonstrate that compared to the state-of-the-art collective intelligence baseline system MiroFish, NEXUS achieves significant improvements across key metrics. The framework provides a systematic solution for situation awareness, evolution forecasting, and response strategy generation in complex public opinion scenarios.

Keywords: *Public Opinion Analysis, Multi-Agent Simulation, Knowledge Graph, Large Language Model, Social Computing*

1. Introduction

In today's digital society, public opinion events increasingly exhibit characteristics of cross-platform dissemination, fragmentation, high noise, and rapid evolution. Their formation and transformation are collectively shaped by the complex interactions among multiple stakeholders. While existing LLM-based public opinion analysis systems can generate information summaries, they still face significant limitations in aligning with domain knowledge, constructing traceable evidence chains, and simulating the dynamic evolution of public opinion [1]. This underscores the need to move beyond traditional static statistical forecasting and develop simulation and inference systems capable of realistically capturing multi-agent interactions.

To systematically address these challenges, this paper proposes an integrated framework named **NEXUS**. Centered on the concept of transforming fragmented information streams into a simulatable knowledge system, the framework organically integrates four core modules: domain-adaptive fine-tuning, user-anchored deep search, semantic graph-based knowledge structuring, and multi-agent virtual society simulation, thereby constructing an end-to-end analytical and simulation pipeline.

Specifically, the contributions of this framework can be summarized as follows:

- We propose a domain-adaptive large language model training and knowledge injection mechanism, constructing a domain knowledge-enhanced framework for public opinion simulation tasks.
- We design a user-anchored deep search and entity–relation extraction module to achieve precise alignment with user intent and structured modeling of relationships.
- We implement a large-scale multi-agent virtual society simulation platform that supports dynamic simulation of heterogeneous agent interactions, opinion diffusion, and evolutionary processes.

2. Related work

Domain-Adapted Fine-Tuned LLM. Recent studies have shown that general-purpose LLMs often suffer from domain shift when applied to specialized analytical scenarios, leading to terminology misuse, contextual inconsistency, and unstable reasoning behaviors [2]. To mitigate these limitations, recent research has explored domain-adaptive fine-tuning and parameter-efficient adaptation techniques. LoRA introduces a lightweight low-rank adaptation mechanism that significantly reduces the computational cost of fine-tuning while preserving the general capabilities of the backbone model [3]. Subsequent frameworks such as LLaMA-Factory further improve the scalability and practicality of large-scale domain adaptation pipelines [4]. In addition, prior reasoning methods such as KAVER enhance factual consistency and reasoning traceability [5].To enhance terminology alignment, reasoning robustness, and overall analytical consistency, our framework needs to incorporate domain-adaptive fine-tuning capabilities.

Deep Search & Retrieval Agents. Recent work on web-based deep search and multi-agent information retrieval has improved large language models' ability to perform complex information acquisition through tool use, search engine interaction, and reasoning agents. Toolformer enables autonomous tool invocation via self-supervised learning, improving external information access [6]. Reflexion introduces self-reflective feedback loops that iteratively enhance reasoning over multi-turn interactions [7]. WebGPT further leverages reinforcement learning from human feedback to enable web browsing and evidence aggregation [8]. How-

ever, they still lack adaptive control over search depth and exploration strategies. They also rarely incorporate user-specific attention modeling, making it difficult to align retrieval and reasoning with personalized analytical perspectives. This limitation is especially critical in public opinion analysis.

Entity–Relation Structuring. Recent graph-memory and graph-retrieval studies have explored how extracted semantic entities can support structured LLM reasoning. AriGraph links episodic memory with semantic entity structures for long-term agent memory [9], Triplex improves entity and relation triple extraction from unstructured text [10], and LightRAG combines local entity-level information with global graph structures to improve retrieval efficiency [11]. However, AriGraph mainly focuses on memory organization, Triplex on extraction quality, and LightRAG on retrieval efficiency, while public opinion simulation requires temporally aware, query-aligned, and traceable event structuring. Therefore, we adopt Zep as the entity–relation structuring component of NEXUS, as its temporal knowledge graph architecture supports entity resolution, relationship organization, temporal updates, and graph-based retrieval.

Analysis and Prediction of Public Opinion. Existing studies have explored public opinion analysis and prediction from different data-driven perspectives. One study constructs a public opinion crisis prediction model based on social network mining, using CNN and LSTM techniques to extract textual and temporal features for crisis trend prediction [12]. Another uses LLMs to estimate public opinions from demographic and information [13], while a third evaluates whether multiple LLMs can track favorability trends in the 2024 U.S. presidential election [14]. However, these approaches mainly predict aggregate attitudes or crisis trends, often functioning as black-box predictors with limited interpretability. They also provide limited modeling of stakeholder interactions, platform-mediated diffusion, and emergent opinion dynamics. Therefore, it is necessary to move beyond aggregate prediction and introduce multi-agent virtual world simulation, where heterogeneous agents can interact through social actions and platform mechanisms to support opinion evolution forecasting and response strategy recommendation.

3. Methodology

3.1. Domain-Adapted Fine-Tuned LLM

3.1.1. Collaborative Requirements

Public opinion analysis is highly domain-dependent in both terminology and contextual interpretation, as domains such as politics, academia, and entertainment differ substantially in their conceptual frameworks, linguistic conventions, and patterns of event evolution. General-Purpose LLM often exhibit term misuse and contextual misalignment in cross-domain scenarios, which can compromise the consistency and reliability of analytical outputs[15]. This limitation motivates the need for domain adaptation to ensure alignment with domain-specific knowledge boundaries.

Moreover, opinion simulation require transparent and verifiable reasoning, including explicit evidence such as historical analogy and domain constraints, beyond final predictions.Domain-adaptive fine-tuning can embed these analytical patterns into structured reasoning behaviors[5].

Consequently, incorporating domain knowledge enhances initial output quality and reduces human correction overhead, enabling more efficient and reliable human–AI collaboration focused on higher-level analysis and decision-making.

3.1.2. Problem Decomposition

To ensure high credibility of the knowledge injected into the model, we establish a strict quality assurance framework at the source. All knowledge inputs are required to undergo review and validation by experts before being integrated into the system. The system supports multiple heterogeneous forms of knowledge representation, including structured knowledge graphs and data files, semi-structured regulatory documents and case repositories, as well as unstructured raw textual data. Through this design, we improve the overall reliability and quality of the knowledge base, thereby providing a solid knowledge and reasoning foundation for downstream reasoning and public opinion simulation tasks.

After incorporating domain prior knowledge and addressing the heterogeneity of input data, we build an automated pipeline based on the Easy Dataset framework to efficiently transform unstructured documents into high-quality supervised fine-tuning data. The pipeline first performs multi-format parsing and text chunking to convert documents into structured textual units. It then generates question–answer pairs using an LLM-based approach, incorporating a "genre–audience" configuration and persona-guided prompting to improve task relevance and diversity. Finally, knowledge-enhanced prompting is applied to produce fact-consistent augmented question–answer pairs[16]. All data are organized into a unified standardized format to support seamless integration with model training workflows.

We then adopt a Parameter-Efficient Fine-Tuning (PEFT) paradigm for domain adaptation while preserving the general capabilities of the base model. We construct high-quality question–answer pairs using Easy Dataset[16] and employ them as supervised fine-tuning data, which are then fed into the LLaMA Factory framework for training, resulting in a domain-adapted fine-tuned LLM. This approach leverages lightweight adaptation techniques such as Low-Rank Adaptation (LoRA), which update only a small

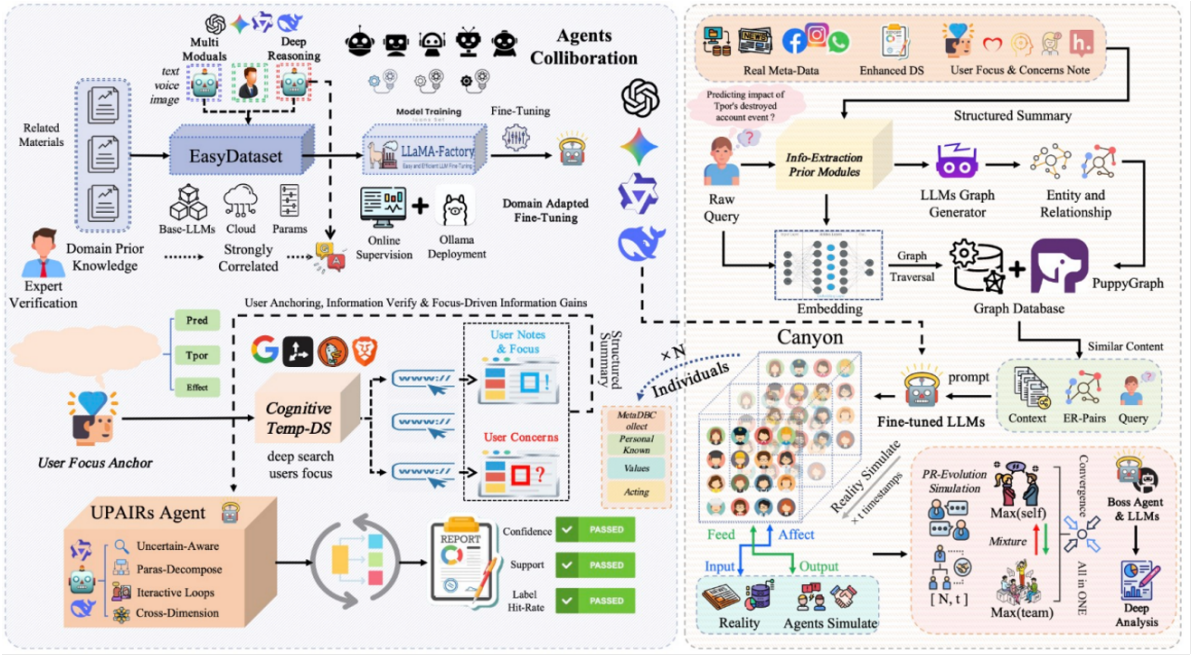


Figure 1. Overview of the NEXUS framework

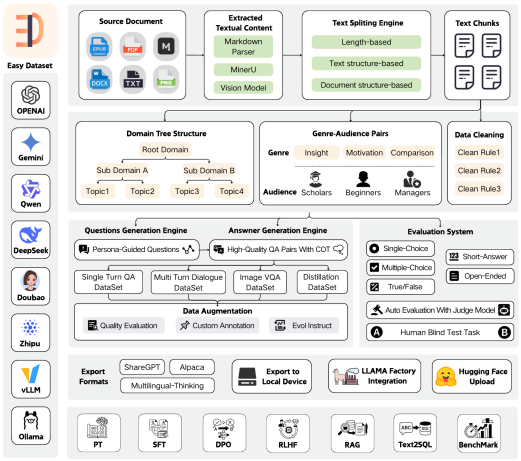


Figure 2. Overview of Easy Dataset framework, adapted from [16].

subset of parameters to inject domain-specific knowledge. This design significantly reduces computational overhead while mitigating the risk of catastrophic forgetting associated with full fine-tuning [3, 4].

Based on the above modular design, we develop a domain-adaptive LLM training and knowledge injection framework for public opinion simulation tasks, enabling an end-to-end pipeline that integrates domain knowledge construction, automated data generation, and parameter-efficient fine-tuning.

3.2. Customized Deep Search Agents

3.2.1. Collaborative Requirements

Considering that different individuals and organizations often develop distinct attention anchors when confronted with the same event, the system should provide personalized public opinion analysis, situation simulation and forecasting, as well as public relations strategy recommendations. To achieve this objective, the system first needs to identify and model users' specific attention anchors, thereby aligning them with their actual information needs and cognitive preferences. This provides clear semantic constraints and directional guidance for subsequent in-depth analysis and multi-agent collaborative reasoning.

In addition, given the widespread presence of noise, incomplete information, and conflicting evidence in real-world information environments, the system must perform deep web retrieval, credibility assessment, and information quality quantification. This is necessary to overcome the limitations of fixed-depth reasoning in complex scenarios, thereby ensuring that the analytical outputs are both reliable and actionable.

Therefore, the system requires two core modules to function: a Deep Search Module for high-quality information retrieval and a User Anchor Analysis Module.

3.2.2. Problem Decomposition

CognitiveTemp Deep Search Agents

To enable deep exploration of web-based information, this study proposes the CognitiveTemp-DeepSearch-

Agents framework, whose core operating mechanism is temperature-driven control. The system takes as input event-related keywords from the user query, a selected search time horizon, configurable search engines (e.g., Google, Baidu), as well as specified content filtering constraints and prioritization-based verification strategies.

Table 1. Multi-agents and their core mechanism in CognitiveTemp DeepSearch Agents and UPAIRS-Agents

Agent	Core Mechanism
Planner	Breaks down complex problems into subproblems
Retriever	Performs multi-query search via Tavily[17] with domain diversification and filtering
Summarizer	Fuses information, classifies evidence, and produces structured initial summaries
Reflector	5-dimensional assessment (perspective, evidence, bias, completeness, contradiction)
Debate	Iteratively refines content via additional retrieval and reflection-driven improvement
Uncertainty	Computes global uncertainty using a 5-factor weighted model to decide continuation
Synthesis	Integrates refined sections into a structured Markdown report with synthesis
Controller	Orchestrates the entire workflow and controlls report generation.

During the information retrieval and evaluation stage, the framework invokes four core agents: the Planner, Retriever, Evaluator, and Reflector. The Planner decomposes complex queries into a set of mutually exclusive and collectively exhaustive subproblems. The Retriever then executes parallel search operations based on this decomposition, calling external search engines to collect web documents relevant to each subproblem.

Subsequently, the Evaluator and Reflector work in a coordinated manner to perform in-depth analysis and quality assessment of each retrieved webpage. They generate a detailed evaluation report in which each document is annotated with key reliability indicators, including confidence score, support level, and support ratio, to quantify both the credibility and relevance of the information.

Finally, the high-quality information selected through this round of intelligent evaluation and filtering is passed to the subsequent user-anchor analysis and evidence-chain construction module, forming the foundational data for generating a structured, in-depth analytical report.

User Anchor Analysis

In the user anchor analysis module, we propose an enhanced real-time information capture mechanism that integrates interactive annotation functionality directly into web-based content interfaces. Specifically, the system employs the Hypothesis API [18]to continuously monitor and record user interaction behaviors over one or multiple documents retrieved during the preceding deep search stage.

When a user highlights a segment of text, the system treats this selection as an explicit attention anchor and immediately captures the corresponding content. In addition, the framework supports flexible natural language annotations over either the selected span or the full document, including commentary, uncertainty marking, and opinion expression. These interaction traces are systematically transformed into structured representations, thereby providing a high-quality, real-time signal for downstream user interest modeling and preference inference tasks.

Table 2. Multi-agents for user anchor deep analysis

	Role	Responsibility	Input	Output
A	Focus Signal Analyst	Extracts trend signals, evolutionary trajectories, and attention intensity from user focus data	User attention data	Trend signal report
B	Content–Evidence Evaluator	Identifies technological evolution evidence, temporal structures, and key milestones from page content	Web content	Technical evolution evidence report
C	Event–Trend Reasoner	Infers technology diffusion paths and industry adoption dynamics from correlated events	Related event information	Event–trend correlation report
D	Trend Strategy Integrator	Integrates outputs from Agents A, B, and C to generate a strategic analytical report for trend forecasting	Analytical outputs (A + B + C)	High-quality strategic report

Based on structured user interaction and annotation data extracted via the Hypothesis API[18], we construct a closed-loop deep analytical generation workflow, incorporating the agents in Table 2. The system takes real-time user attention signals and annotation behaviors as primary inputs and, through a multi-agent hierarchical coordination mechanism, sequentially performs trend signal mining, extraction of technological evolution evidence, and inference of inter-event relationships. It subsequently integrates and synthesizes the derived insights to produce a strategic analysis report characterized by high relevance, high timeliness, high fidelity, and strong personalization.

Beyond responding to explicit user requirements, the report further infers latent intent from behavioral signals and establishes explicit observation anchors and reasoning pivots to support downstream discourse evolution, including opinion dynamics forecasting and scenario simulation.

UPAIRS Agents

CognitiveTemp-DeepSearch-Agent regulates the breadth and creativity of web retrieval. To achieve high factual fidelity and strong logical robustness in long-form deep synthesis, we further propose an innovative Pressure-Driven paragraph-level iterative reflection search architecture, UPARIs Agent. The core of this framework lies in a five-factor uncertainty quantification model, which continuously computes an uncertainty score across five dimensions: evidence authority, information completeness, perspective diversity, content consistency, and temporal relevance, thereby serving as the driving force of system operation.

The framework coordinates the 8 agents in Table 1 and adaptively allocates retrieval budget and reasoning depth according to uncertainty thresholds. It further incorporates

mechanisms such as semantic entropy estimation, variance penalization, diversity-aware re-ranking, and cross-dimensional confidence calibration to perform multi-level refinement of retrieved results.

This design establishes a self-evaluating and iteratively optimizing search–reasoning system, effectively mitigating information conflicts and hallucinations in RAG systems.

3.3. Semantic-Graph Knowledge Structuring

3.3.1. Collaborative Requirements

After the Deep Search Module and User Anchor Analysis Module identify the user's attention anchors and generate a user-focused report, the system still needs to transform this information into a knowledge foundation for subsequent multi-agent simulation. The purpose of this module is to reorganize the raw user query and the user-focused report into two complementary forms of knowledge: semantic embedding contexts and structured entity-relation graphs.

Considering that different agents in the later simulation need to reason from different perspectives, the system must preserve both contextual relevance and structural traceability. Contextual relevance ensures that the retrieved information remains aligned with the user's current query and attention focus, while structural traceability ensures that key actors, platforms, actions, responsibilities, and causal relations within the event can be explicitly represented and retrieved.

This motivates the design of a semantic-graph knowledge structuring module, which bridges user-specific concerns with simulation-ready knowledge. By combining semantic contexts with entity-relation structures, this module reduces fragmented reasoning and provides a reliable, query-aligned, and interpretable foundation for subsequent multi-agent simulation.

3.3.2. Problem Decomposition

To construct a knowledge foundation for subsequent multi-agent simulation, this module decomposes the problem into two core sub-problems: semantic context construction and entity–relation structuring. The former focuses on transforming the raw user query and the user-focused report into retrievable embedding-based contexts, while the latter focuses on extracting structured entities and relations and storing them in a graph database.

Semantic Context Construction

Semantic context construction focuses on aligning the raw user query with the user-focused report through embedding-based representation. The raw query represents the user's explicit information need, while the user-focused report provides richer contextual information, including attention anchors, issue scope, and analytical direction. To connect these two forms of input, the system converts the relevant textual content into dense embedding vectors.

After embedding generation, cosine similarity is used to measure the semantic relevance between the raw query embedding and the embeddings derived from the user-focused report. Since dense text embeddings mainly represent semantic direction, cosine similarity can identify contexts that are conceptually related to the user's query, even when the wording is different.

The generated embeddings are stored in FAISS, which supports efficient similarity search over high-dimensional vector representations [19]. During retrieval, the user's query is encoded into an embedding vector and compared with the stored embeddings in FAISS. The system then returns the top-ranked semantically relevant contexts. The output of this stage is the retrieved embeddings context, which provides query-aligned contextual evidence for the subsequent reasoning process.

Entity–Relation Structuring

This stage preserves the structural relationships behind the user-focused information. Public opinion events are not only composed of textual descriptions, but also involve key actors, platforms, actions, responsibilities, and causal links. Therefore, the system needs to extract entity–relation structures from the same user-focused information.

To address this problem, the system uses Zep to extract key entities and relations from the textual content. Zep provides a temporal knowledge graph architecture for agent memory, which supports the organization of information into structured entity–relation representations [20]. The extracted graph structure is then stored in PuppyGraph, which functions as the graph storage and querying layer [21].

During graph retrieval, the system uses the entities or metadata associated with the retrieved semantic contexts to locate corresponding nodes and relations in PuppyGraph. This allows the system to retrieve not only semantically relevant text, but also the underlying event structure. For example, in a public opinion event, the graph retrieval process may return related stakeholders, platform responses, responsibility links, and causal relations.

As a result, this module produces three core outputs: the user query, the retrieved embeddings context, and the related nodes and relations. Together, these outputs form a query-aligned and structurally traceable knowledge package, providing both semantic context and explicit event relationships for subsequent multi-agent simulation and reasoning.

3.4. Multi-agent Virtual World Simulation

3.4.1. Collaborative Requirements

Public opinion analysis is a complex process of social cognition, fundamentally characterized by the continuous evolution of diverse viewpoints, deep group interactions, and the dynamic diffusion of information.

We need to perform virtual world simulations, construct-

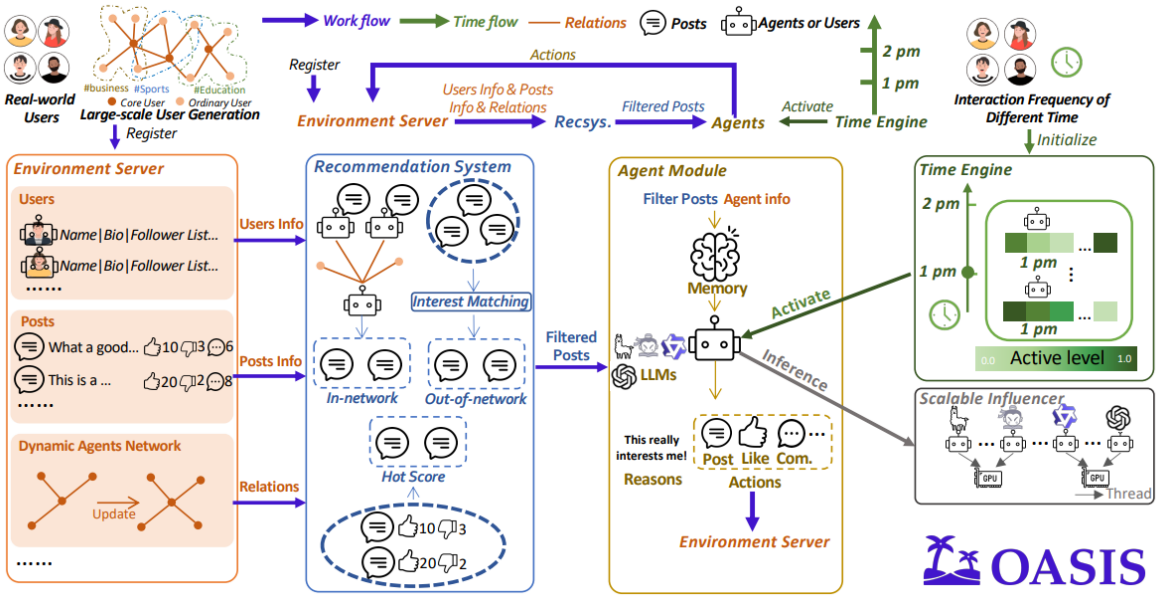


Figure 3. The workflow of OASIS

ing agent populations that are deeply mapped to real-world social structures. By simulating dynamics such as opinion confrontation, emotional contagion, and polarization evolution, we can achieve virtual exploration of how public opinion develops over time.

During this process, individuals and groups may undergo attitude shifts, belief reinforcement, or stance reversals through interaction. Therefore, it is necessary to construct sufficiently large and diverse agent populations to reproduce realistic information ecosystems and capture social processes such as value convergence and opinion polarization.

As mentioned earlier, we have identified key entities and their interrelationships. However, to achieve higher-fidelity social simulations, these groups must be enriched along personality dimensions, enhancing agents' behavioral diversity and psychological realism.

From a client-oriented perspective, the system should not only provide public opinion analysis but also support decision-making, generating actionable public relations strategies based on simulation outcomes. Additionally, for scenarios where clients may wish to focus on specific stakeholder perspectives or gain multi-angle insights, the system should further support in-depth exploration of entity-level viewpoints.

3.4.2. Problem Decomposition

In the earlier stage, we obtained the knowledge graph (KG) representing the entities and their relationships involved in the event.

Next, we refine and simulate these entities based on analysis requirements. To achieve this, we extend the personality dimensions of each agent by introducing cognitive preference differences, behavioral habits, and social influence parameters to construct detailed Personal Profiles. We also define users' online behaviors, such as posting frequency, commenting activity, and engagement levels. This process results in agents with distinct MBTI types, event experiences, behavior patterns, unique memories, and social networks, forming a comprehensive Knowledge Graph.

All of this information is then fed into the Domain-Adapted Fine-Tuned LLM from Section 3.1. Leveraging domain-specific knowledge, the LLM ensures that all agent-generated interactions are consistent with professional knowledge, preventing errors that would conflict with domain facts, thus providing higher accuracy and consistency in the simulation.

On the OASIS platform in Figure 3, we can create up to one million agents, each capable of performing rich social actions such as posting, commenting, liking, disliking, searching content, following, or muting other users [22]. Individual behaviors reflect personal preferences, while collective interactions generate complex social dynamics, including information propagation, group polarization, opinion formation, and herd behavior. The system also integrates recommendation mechanisms based on interests or popularity, simulating how users discover and interact with content, making the simulation more realistic.

The outcome of the simulation produces an Analytical Report for opinion analysis and decision support. To uti-

lize the Analytical Report, we have designed a Report QA Agent that provides both opinion analysis and public relations strategy recommendations (Basic or Advanced modes, configurable via prompt engineering).

For multi-perspective insights, we also implement a Simulated Interview module, allowing one-on-one interactions with different entities to explore their viewpoints in depth. Additionally, we offer both an Official version and a Heart-to-Heart version, with the former for formal analysis and the latter to simulate the entities' genuine internal thoughts, making the analysis both engaging and realistic.

4. Evaluation & Discussion

4.1. Experimental Setup

4.1.1. Implementation Details

This study adopts a hybrid cloud–local computing architecture. Domain-adaptive fine-tuning is performed in the cloud using the AutoDL platform with pre-configured environments and automated pipelines. Depending on model scale, different GPU instances (e.g., RTX 4090, A40, A800/H100) are selected to balance computational efficiency and cost. The fine-tuned models are quantized using toolchains such as Ollama [23] and deployed locally for low-latency inference on devices equipped with Apple M-series chips or NVIDIA RTX GPUs.

For multi-agent virtual world simulation, high-performance computing nodes with large GPU and system memory are required to support concurrent interactions, real-time knowledge graph updates, and complex reasoning tasks. Detailed configurations are provided in the appendix.

4.1.2. Baseline

This study adopts MiroFish as the primary baseline. We reproduce its full framework and further implement selected key modules as controlled variants for ablation-style comparison. MiroFish is an open-source large-scale multi-agent coordination system designed for LLM-based collective intelligence and group-level prediction tasks. It is built upon a decentralized architecture in which multiple autonomous agents interact and exchange information, enabling emergent collective reasoning capabilities [24].

The selection of MiroFish is motivated by its strong alignment with the proposed method in both design and functionality. Both approaches rely on decentralized multi-agent coordination mechanisms to perform information integration and analysis in complex, large-scale environments, particularly for tasks involving information aggregation and opinion or narrative simulation[24]. This alignment makes MiroFish a suitable and representative baseline for evaluating improvements in report structure design and reasoning stability.

To ensure experimental fairness, Mirofish and NEXUS are evaluated under identical task settings, external retrieval constraints, and generation budgets, ensuring that performance differences primarily reflect methodological variations rather than differences in resources or experimental conditions.

4.1.3. Evaluation Metrics

Table 3. Notation used in evaluation metrics

Symbol	Description	Symbol	Description
EIS	Overall score	R, K, M, S, I	Five main qualities
s	Source relevance	r	Retrieval signal
e	Evidence quality	d	Evidence density
sch	Schema completeness	ent	Entity coverage
rel	Relation signal	$dens$	Graph density
$conn$	Connectivity	rea	Reasoning signal
D	Diversity fit	A	Agent agreement
p	Progress	b	Balance
i	Intensity	dn	Density
a	Adaptability	m	Memory signal
x	Interaction	n	Novelty
g	Grounding	st	Structure quality
ev	Evidence quality	act	Actionability
ali	Alignment	coh	Coherence
pro	Process quality		

This study proposes a hierarchical automated evaluation framework to measure system performance across retrieval, knowledge graph construction, multi-agent collaboration, simulation capability, and insight generation. The goal is to enable stable and reproducible quantitative evaluation in the absence of human annotations by relying on computable proxy signals extracted from system outputs and runtime logs. The final overall score is defined as:

$$EIS = w_r R + w_k K + w_m M + w_s S + w_i I \tag{1}$$

Retrieval Quality is used to measure the effectiveness of information acquisition and evidence support, calculated as a weighted combination of source relevance, evidence density, and coverage:

$$R = \alpha_1 s + \alpha_2 r + \alpha_3 e + \alpha_4 d \tag{2}$$

KG quality is defined for evaluating structured knowledge construction and reasoning capabilities:

$$\begin{aligned} K = {} & \beta_1 sch + \beta_2 ent + \beta_3 rel \\ & + \beta_4 dens + \beta_5 conn + \beta_6 rea \end{aligned} \tag{3}$$

Multi-agent quality is for measuring the information diversity, degree of divergence, and convergence in collaboration among different agents.:

$$M = \gamma_1 D + \gamma_2 A \tag{4}$$

Simulation quality is used to evaluate the stability and adaptability of behaviors in dynamic environments:

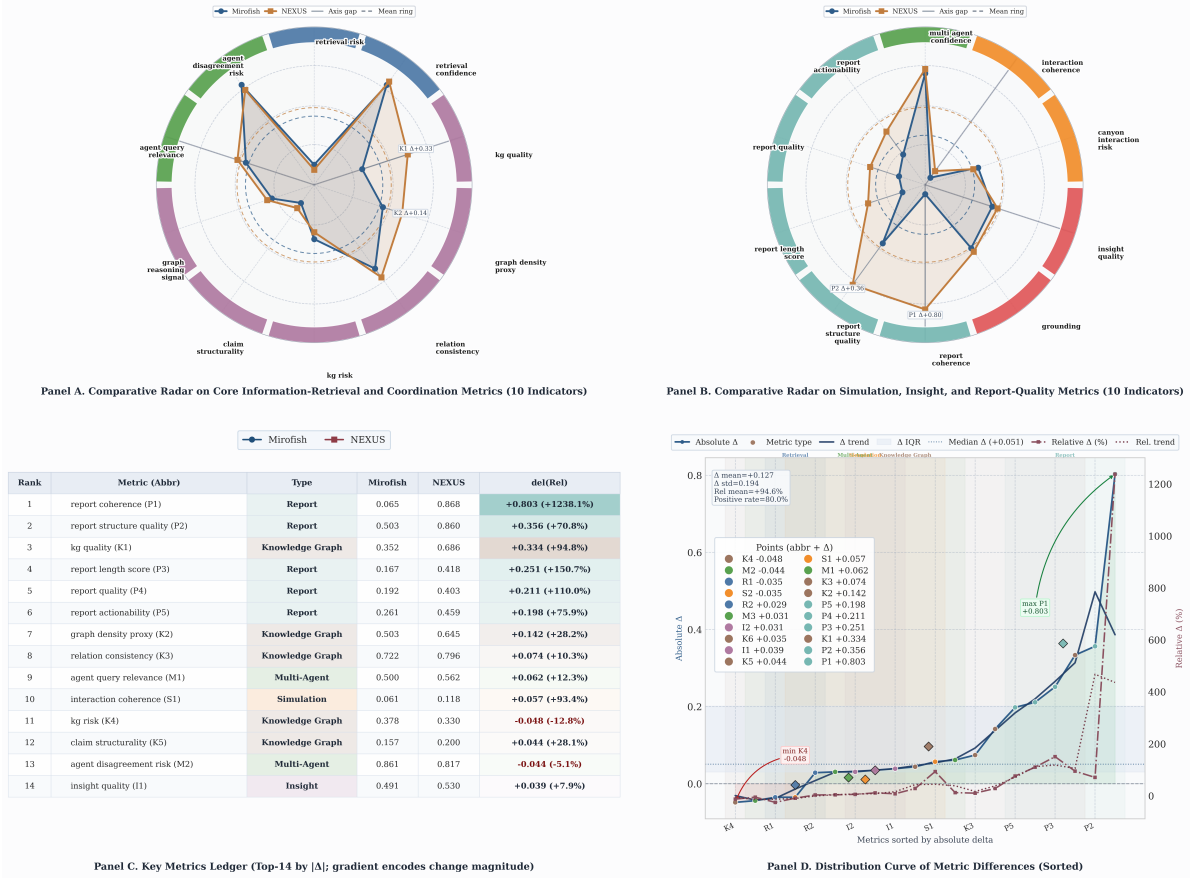


Figure 4. Mirofish vs. NEXUS Multi-Dimensional Core Capability Assessment: A Comparative Evaluation of Information Retrieval, Coordination, Simulation, and Report Quality (Radar Charts, Metrics Ledger, and Difference Distribution)

$$S = \delta_1 p + \delta_2 b + \delta_3 i + \delta_4 dn + \delta_5 a + \delta_6 m + \delta_7 \tag{5}$$

Insight quality is used to evaluate the semantic quality and factual support level of the final analytical output:

$$I = \lambda_1 I_{ag} + \lambda_2 R_{rep} \tag{6}$$

where:

$$I_{ag} = \mu_1 n + \mu_2 rel + \mu_3 g \tag{7}$$

$$R_{rep} = \nu_1 st + \nu_2 ev + \nu_3 act + \nu_4 ali + \nu_5 coh + \nu_6 pro \tag{8}$$

Finally, system reliability is measured via a risk function, serving as the dual form of the quality metric.:

$$risk = \rho_1 R_r + \rho_2 K_r + \rho_3 M_r + \rho_4 (1 - S) + \rho_5 I_r \tag{9}$$

4.2. Performance Comparision & Discussion

We compare NEXUS with the baseline model Mirofish. As shown in Figure 4, NEXUS demonstrates clear and consistent superiority across most evaluation dimensions.

In the radar chart of Figure 4, the blue-green region represents the overall Report evaluation metrics. The dark blue region corresponds to Retrieval Quality (R), purple denotes KG Quality (K), yellow-green indicates Multi-agent Quality (M), orange represents Simulation Quality (S), and red corresponds to Insight Quality (I).

NEXUS significantly outperforms Mirofish in most core aspects, including report quality, simulation interaction consistency, knowledge graph construction, and multi-agent coordination. In particular, NEXUS achieves substantial gains in report quality, where metrics such as coherence, structural quality, length adequacy, and readability show improvements exceeding 70%, with some indicators exhibiting order-of-magnitude gains. This reflects a substantial enhancement in structured information organization, logical consistency, and overall readability.

Although NEXUS performs strongly across most met-

rics, Mirofish slightly outperforms NEXUS in a few risk-related indicators, including knowledge graph risk control and multi-agent disagreement risk. This suggests that Mirofish adopts a more conservative or stability-oriented strategy in handling uncertainty and coordination among agents. While NEXUS achieves overall performance gains, these results highlight the importance of maintaining a balance between performance improvement and robustness in risk-sensitive settings.

5. Future Work

To build a more responsible, professional, and practically effective decision-support system, the future evolution of the NEXUS framework will focus on three closely interlinked dimensions.

First, an embedded mechanism for ethical alignment and bias mitigation must be developed to ensure that the outputs of the simulation process adhere to ethical guidelines.

Second, to overcome the current limitations in the depth and traceability of decision recommendations, a structured "strategy-outcome" decision knowledge base should be constructed. This base will integrate historical cases, expert knowledge, and simulation data to form a reasoning-capable domain-specific knowledge graph, thereby supporting evidence-based judgment and interpretable decision-making.

Third, to close the loop from analytical prediction to strategy optimization, the system will re-inject proposed strategies into the multi-agent simulation environment, initiating a new round of "strategy release – public feedback" deduction. By observing the chain reactions and evolutionary pathways triggered by the strategy in the virtual society, the system will be able to prospectively evaluate the real-world effectiveness and potential risks of the adopted strategies.

6. Conclusion

In conclusion, we design NEXUS, a multi-agent framework for public opinion analysis and social simulation. Our framework integrates four core components: domain-adapted fine-tuned large language models, deep search and user anchor analysis, semantic knowledge graph construction, and multi-agent virtual social simulation, forming a unified end-to-end analytical pipeline.

Experimental results show that, compared to the state-of-the-art collective intelligence baseline system MiroFish, NEXUS demonstrates significant advantages across multiple core dimensions, including retrieval quality, knowledge graph completeness, multi-agent collaboration efficiency, simulation stability, and structural quality, logical consistency, and actionability of the final report. This validates the overall effectiveness and technical advancement of the proposed framework in complex public opinion analysis tasks.

7. Contribution

Qianqi Chen(2330034001): Responsible for organizing the overall project architecture and confirming the detailed logic of the overall codebase. Proposed improvement ideas such as the public opinion decision-making agent, led the production of the final presentation slides, designed the detailed presentation logic, coordinated the paper structure, and supervised the division of work among team members.(***20%***)

Minghao Li(2330034027): Responsible for the development and implementation of the system's core code. Built the code architecture for all four core modules of the NEXUS framework, completed evaluation visualization and ablation study, designed the GitHub project page for project demonstration, and provided comprehensive technical support to the team.(***30%***)

Yirui Liang(2330034030): Responsible for the core writing, polishing, and finalization of the paper. Proposed an important improvement idea of forming a closed loop from analysis and prediction to strategy optimization and evaluation. Also coordinated the checking of the technical environment configuration, aligned the logic of the paper and the presentation, and completed the PPT production, idea integration, and team coordination for the project pre-presentation.(***20%***)

Chuqing Hou(2330034014): Responsible for reviewing the code details of the domain fine-tuning module. Developed a detailed understanding of the relevant technical and mathematical principles, provided suggestions for improving this part of the code, and assisted in writing and refining the presentation content and slides.(***10%***)

Guangmei Li(2330034025): Responsible for literature research and content organization of similar projects. Organized the content of the user anchor module and assisted in writing and refining the presentation content and slides.(***10%***)

Yufei Li(2330025049): Responsible for organizing the code and content of the deep search module and assisted in writing the presentation content.(***10%***)

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NEXUS: Navigating Emergent X-agent Universe Simulator with Unprecedented Insight

Supplementary Material

A. Experiment Setting

System-level Runtime Configuration defines the backend environment, LLM interface, knowledge-memory access, document processing, and social simulation dependencies. These settings provide the shared execution environment for the downstream search, graph structuring, and simulation modules.

Table 4. System-level runtime configuration

Item	**Setting**
Runtime	Python 3.11+
Backend API	Flask, Flask-CORS
LLM interface	OpenAI-compatible API
Default backend LLM	`gpt-4o-mini`
LLM timeout	180s
Connection timeout	30s
Knowledge memory	Zep Cloud 3.13.0
Social simulation	CAMEL-OASIS 0.2.5
Document parsing	PyMuPDF
Data validation	Pydantic

B. Ablation Study

Metric	**Without**	**With**	**Improvement (Δ)**
Step 3 EIS	0.578	0.668	+0.090
KG Quality	0.352	0.671	+0.319
Retrieval Quality	0.862	0.875	+0.013
Multi-Agent Quality	0.670	0.686	+0.016
Simulation Quality	0.625	0.638	+0.013
Insight Quality	0.511	0.523	+0.012
Overall Risk	0.331	0.312	-0.019

Table 5. Impact of Prior Knowledge on EIS Evaluation Metrics

Table 6. Metrics Showing Minor Regression

Metric	Direction	No Markdown	With Markdown	Regression
novelty	higher	0.931598	0.885614	-0.045984
insight_quality_agentd	higher	0.636847	0.623972	-0.012876
report_coherence	higher	0.882046	0.876001	-0.006044

C. Case Study

Our framework is not only applicable to public opinion analysis, simulation, prediction, and decision-making suggestions for public opinion events on social media, but can also be utilized for hotspot analysis and future trend assessment in specialized fields such as academia. Next, we will elaborate on this through two examples: the Qingyu Tpor Incident (public incident in entertainment) and GraphRAG Trend.

Table 7. Aggregated Performance Gains Across Themes

Theme	#Metrics	Positive	Negative	Net Gain	Avg Gain
knowledge_graph	24	9	0	+1.038722	+0.043280
multi_agent	23	11	0	+0.797378	+0.034669
simulation	17	7	0	+0.442856	+0.026050
insight_report	18	8	2	+0.602136	+0.033452

Table 8. Quality Gate Evaluation Comparison

Gate	No Markdown	With Markdown
insight_ge_0_60	✗	✓
kg_ge_0_60	✗	✓
multi_agent_ge_0_55	✗	✓
overall_eis_ge_0_70	✗	✗
retrieval_ge_0_75	✓	✓
simulation_ge_0_80	✗	✗

C.1. Public Incident in Entertainment

During streamer Qingyu Tpor's live broadcast of Delta Action, citing the expiration of the supporter's "Captain" monthly membership as the reason, he logged into and wiped out the game account of long-term supporter without prior notice—exploiting the live stream delay to complete the action within 30 seconds. The account was valued at over ten thousand yuan, including rare items and top-tier gear. The incident quickly sparked widespread condemnation across the internet and stirred heated discussions in mainstream media.Within this 48-hour window, we conducted public opinion simulation and successfully predicted the subsequent trend. The following outlines our operational process in Figures 5–11.

C.2. Trend Prediction in Academia

To demonstrate the applicability of NEXUS beyond crisis-oriented social media events, we use GraphRAG trend analysis as an academic emerging-hotspot case. GraphRAG represents a fast-evolving research and professional domain, where users may seek to understand unresolved challenges, future development directions, and potential research opportunities. We will illustrate in Figures 12–18.



Figure 5. Complete Knowledge Graph of Qingyu Tpor Incident



Figure 6. Entity Summary of Qingyu Tpor Incident

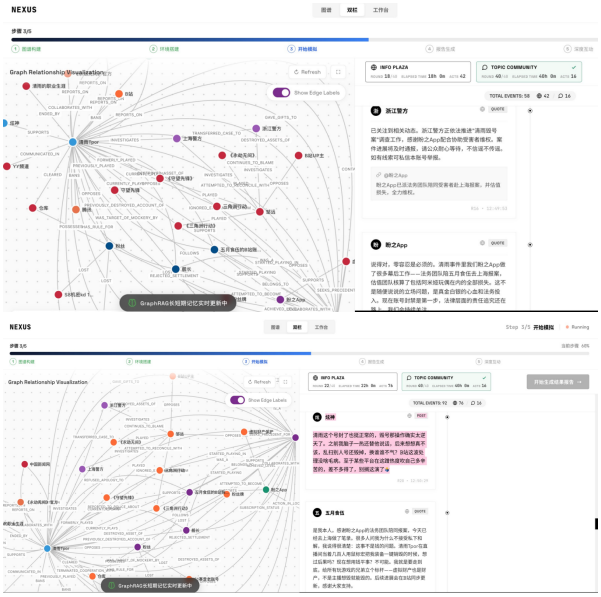


Figure 7. Examples of Multi-Agent Interaction in Qingyu Tpor Incident



Figure 8. Simulated Interview with the Victim in Qingyu Tpor Incident



Figure 9. Official Version vs. Heart-to-Heart in Qingyu Tpor Incident



Figure 10. Analytical Report in Qingyu Tpor Incident



Figure 11. Report Q&A Agent of Qingyu Tpor Incident



Figure 12. Complete Knowledge Graph of GraphRAG Trend



Figure 13. Entity Summary of GraphRAG Trend

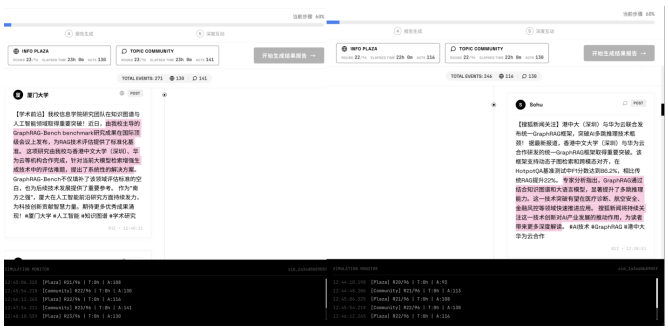


Figure 14. Examples of Multi-Agent Interaction in GraphRAG Trend



Figure 15. Simulated Interview with Researcher in GraphRAG Trend



Figure 16. Official Version V.S. Heart-to-Heart in GraphRAG Trend



Figure 17. Analytical Report of GraphRAG Trendt



Figure 18. Report QA of GraphRAG Trend